AI-Based Adaptive VR Meditation System

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Project Proposal Report

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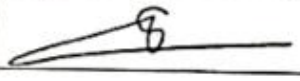
March 2026

DECLARATION

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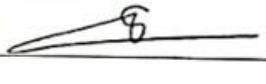
Name	Date	Signature
Mr. Samadhi Rathnayake	19/03/2026	

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ABSTRACT

Psychological stress is a major health concern that requires effective management to prevent its harmful effects on both mental health and physical health. The traditional methods used for stress assessment depend on subjective questionnaires, while modern wearable systems use physiological signals like Heart Rate Variability (HRV) to deliver objective stress monitoring. The current stress detection models face critical restrictions because they cannot operate in various real-world environments and they depend on two specific stress levels and they do not include normal daily biological process changes for human body signals.

The research resolves existing issues through a stress inference model which uses time-dependent Cosinor rhythmometry and machine learning and deep learning methods. The system tracks human body normal HRV patterns throughout the day to create customized health measurement systems and it uses Cosinor residuals to determine actual stress-based changes. The system uses a hybrid method which combines XGBoost for organized feature assessment and CNN-LSTM for learning time-based patterns to achieve multi-level stress assessment.

The project uses multiple technologies which include Python and NeuroKit2 for heart rate variability analysis and CosinorPy for rhythm analysis and PyTorch for deep learning and XGBoost for machine learning. The model development process uses WESAD benchmark datasets for training and assessment and it uses cross-dataset testing along with real-time evaluation methods for validation purposes.

The expected outcome is a robust and accurate multi-level stress detection system capable of operating in real-world environments with improved generalization and reduced false positives. This component contributes to adaptive VR-based meditation systems by enabling real-time personalized interventions.

From a broader perspective, the proposed solution supports digital mental health advancements, improves user well-being, and offers commercialization potential as a scalable stress inference API for healthcare and wellness platforms.

Keywords: Stress Inference, Heart Rate Variability (HRV), Cosinor Rhythmometry, CNN-LSTM, XGBoost

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
CNN	Convolutional Neural Networks
CoEAI	Centre of Excellence for Artificial Intelligence
DSR	Design Science Research
GPU	Graphics Processing Unit
HF	High Frequency
HRV	Heart Rate Variability
IoT	Internet of Things
LF	Low Frequency
LSTM	Long Short-Term Memory (networks)
MESOR	Midline Estimating Statistic of Rhythm
MQTT	Message Queuing Telemetry Transport
PSD	Power Spectral Density
REST	Representational State Transfer
RMSSD	Root Mean Square of Successive Differences
ROC	Receiver Operating Characteristic
RR	R-R Interval (The time between two consecutive heartbeats)
SDG	Sustainable Development Goal
SDNN	Standard Deviation of NN (Normal-to-Normal) intervals
SWELL-KW	SWELL Knowledge Work Dataset
TILES	Tracking Individual Performance with Sensors
VR	Virtual Reality
WESAD	Wearable Stress and Affect Detection (dataset)
XGBoost	Extreme Gradient Boosting

Table 1: List of abbreviations

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1. INTRODUCTION

1.1 Background and Context

Psychological stress represents a profound real-world health challenge, inherently subjective in its perception yet capable of triggering severe physiological and behavioral consequences if left unmanaged [1]. While historical assessments relied heavily on subjective questionnaires, there is now a critical need for continuous, objective monitoring in digital health using unobtrusive wearable technology [1]. However, contemporary stress detection systems frequently suffer from a fundamental "context-blindness" because they ignore the natural, continuous fluctuations of the autonomic nervous system, evaluating physiological markers in isolation from their temporal reality [2]. Addressing this oversight through temporal awareness is not merely a theoretical benefit but an empirical necessity for accurate physiological modeling [3]. Recent statistical evidence unequivocally demonstrates that heart rate variability and other biological signals are heavily influenced by natural daily cycles; explicitly removing these underlying circadian rhythms before classification can lead to significant accuracy gains of up to 13.67%, proving that robust stress inference must account for time-dependent physiological variations [3].

When evaluating the current landscape of affective computing, the limitations of existing analytical solutions become readily apparent [4]. A primary challenge is the "brittleness" and distinct lack of generalizability in current machine learning models; classifiers trained on highly controlled laboratory data frequently fail to maintain their performance when deployed across varied, real-world datasets due to inter-individual variability and environmental noise [4]. Furthermore, the prevailing reliance on binary "stress vs. no-stress" classifications is demonstrably insufficient for deploying sophisticated therapeutic interventions that require a nuanced understanding of cognitive load, such as differentiating between working under time pressure and facing unexpected email interruptions [5]. Underlying these shortcomings is the widespread use of static physiological thresholds, which entirely fail to account for the longitudinal, trait-like health patterns and day-to-day variances inherent in human physiology [6]. By neglecting the individualized and dynamic nature of these baseline metrics, existing systems are prone to frequent misclassifications, thereby limiting their clinical reliability and operational scope in practical settings [4].

To resolve these critical operational bottlenecks, the proposed solution for ‘Multi-Level Stress Inference via Temporal Pattern Modeling & Mathematical Rhythmometry’ module introduces a highly robust hybrid framework that intrinsically accounts for biological time structures [7]. Specifically, this approach integrates Cosinor Rhythmometry to model the known periodicities of human physiology, such as circadian rhythms allowing the system to establish a personalized, dynamic physiological baseline characterized by individual adjusted means and amplitudes [7]. By subtracting these expected rhythmic variations from the raw data, the system generates isolated

'Cosinor residuals' that represent true stress-induced physiological deviations [7]. Using these residuals as refined input features for sophisticated deep learning architectures dramatically improves diagnostic precision, as advanced neural networks require meticulously transformed and interpretable feature banks to capture complex affective states accurately [8]. Processing these temporal and spectral features through multi-modal convolutional frameworks successfully prevents the loss of critical physiological correlations, establishing a powerful 4-level stress inference engine [9]. Ultimately, this mathematical precision and advanced AI-driven healthcare framework directly aligns with Sustainable Development Goal 3 (Good Health and Well-being) and significantly advances the strategic objectives of the Centre of Excellence for Artificial Intelligence (CoEAI) research cluster.

1.2 Literature Review

The proposed AI-Based Adaptive VR Meditation System functions as an advanced closed-loop system which delivers customized mental health treatment to users. The system begins its operation with employing a unique IoT sensing system to collect accurate physiological data from users. The current system processes data streams through a module which uses advanced multi-level inference techniques to identify psychological stress by analyzing temporal patterns. The system then uses stress indices to control a virtual reality environment that adjusts visual and auditory elements in response to real-time changes. The second validation uses voice stress assessment and cross-modal anomaly detection to validate system performance. The integrated approach enables the system to maintain contextual understanding of user biological and psychological states rather than reacting solely to user actions.

The history of wearable stress detection technology development has depended on basic datasets which include WESAD because this dataset demonstrated that researchers could use multimodal signals which include Heart Rate Variability (HRV) and Electrodermal Activity to identify different emotional states [1]. The 2025 survey by Kyrou et al. [2] shows that the field has moved away from basic binary "stress vs. no stress" detection methods to advanced deep learning models which allow for detailed emotional expression analysis. The current research methods which exist now focus on multi-class classification according to the findings of Mortensen et al. [5] and Hemakom et al. [10] because they believe that psychological stress needs to be divided into specific categories which include relaxed, mild, moderate and high stress for effective clinical or therapeutic purposes.

The current research trends show an important change because researchers now develop frameworks which can understand how time periods affect their results. The traditional models of systems show their inability to function properly because they do not consider the natural 24-hour biological rhythms The study conducted by Shen et al. [3] proved that HRV data analysis needs to track circadian rhythms because this process enables stress evaluation to reach its maximum precision throughout all hours of the day. The mathematical precision of this work depends on

Cosinor-based rhythmometry which Cornelissen [7] developed to estimate changing human physical health patterns. Researches use "Cosinor residuals" to determine actual psychological stress through HRV measurements which show difference between actual findings and expected rhythmic patterns.

The market of current products and services shows a strong preference for consumer wearables which provide users with resting HRV tracking together with long-term health monitoring capabilities [6]. The research work by Chen et al. [8] which was presented at a conference investigated how to improve IoT-based sensors through feature augmentation and signal transformation to enhance their daily operational performance. Although commercial services and research-grade services deliver easy access to their offerings, both service types depend on common population data which reduces their accuracy. The Xiang et al. [9] study shows that modern systems need to combine time features with frequency features through multi-modal deep learning because this method helps to understand how people experience their unique stress patterns.

Technological advancements have achieved progress, yet current systems show multiple critical deficiencies. The main issue centers around the "brittleness" problem which affects current models according to Benchekroun et al. [4] who demonstrate that stress detection models fail to generalize across multiple datasets and different hardware systems. The Kyrou et al. [2] survey shows that existing solutions maintain a "context-blindness" problem because models treat physiological data as isolated moments without understanding how it relates to the user's full daily activities. Quoy et al. [11] solved the problem of deep neural networks needing heavy resources for edge devices by creating efficient compact CNN-based solutions which enable system deployment.

The most serious technical issue in present stress-monitoring services arises from their inability to track the ongoing biological changes which humans experience throughout their daily life cycles. The system misidentifies a normal HRV "dip" which occurs during the afternoon because it fails to recognize that users experience this pattern as part of their daily circadian rhythm. The project combines rhythmometry mathematical foundations from [7] with contemporary deep learning techniques [9], [11] to create a precise and customized inference engine that supports the VR-based therapeutic pipeline.

1.3 Research Gap

Current stress detection systems suffer from severe limitations because they cannot handle multiple real-world situations which extend over time according to their current operational method. The development of high-accuracy models for controlled environments depends on foundational datasets yet these systems lack the ability to function correctly when used with different datasets or actual hardware setups [1], [4]. Research needs to integrate the 24-hour circadian oscillations of the autonomic nervous system which represents temporal biological context into stress inference systems. The existing methods use fixed thresholds based on population data which do not consider

how heart rate variability (HRV) naturally changes during daytime hours, resulting in incorrect diagnoses that mistake natural biological "dips" for psychological stress [3], [6].

Current methods do not meet existing requirements because they depend on binary stress identification which fails to deliver the detailed information needed for advanced medical treatment methods. Deep learning has enhanced classification results, yet most existing systems still use snapshot-based methods which process only distinct data segments without considering the complete time-based medical record of each patient [2][5]. The problem increases because standard machine learning models exhibit "brittleness" which causes them to lose their ability to deliver accurate results when used with different sensors or distinct human body characteristics that need major system adjustments [4][8]. Current consumer wearable devices and Internet of Things services allow users to access their products, yet they do not include the essential mathematical detrending methods which can separate genuine psychological stress signals from the typical biological patterns of daily life [6][9].

The innovative aspect of the proposed solution originates from its creation of a "temporally aware" inference engine which operates through its combined mathematical and artificial intelligence system. The system creates a personalized dynamic physiological baseline through its use of Cosinor-based rhythmometry which changes according to the user’s natural circadian cycle [7]. The method enables researchers to determine "Cosinor residuals" which represent the exact departure of observed HRV from the expected rhythmic baseline and which function as an essential characteristic for the multi-level deep learning classifier [3][9]. The system makes a fundamental change away from conventional black-box systems into a transparent system which uses rhythm-based understanding. The system establishes precision and customization through its ability to separate actual psychological stress from expected biological rhythm patterns which current research and commercial health technology solutions cannot achieve [2][3].

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate a stress inference model that integrates Cosinor rhythmometry and deep learning techniques and assess its performance using classification accuracy and real-time inference metrics within the project duration.

2.2 Specific Objectives

1. **Mathematical Rhythm Modeling:** The Cosinor-based rhythmometry model will be implemented to estimate circadian parameters of Heart Rate Variability (HRV) which include MESOR Amplitude and Acrophase to create a personalized 24-hour physiological baseline.

2. **Circadian Detrending and Feature Engineering:** The design of a signal processing pipeline will enable the extraction of Cosinor residuals which show the HRV difference between actual measurements and the projected circadian basic line. The pipeline will merge these residuals with standard time-domain and frequency-domain HRV measurement methods.

3. **Multi-Level Stress Classification:** The hybrid deep learning model development will create a system which uses CNN-LSTM technology to categorize stress into four levels based on HRV-derived physiological features.

4. **Real-Time Stress Inference:** The development of an efficient inference pipeline will enable processing of physiological data streams which originate from IoT sensors while producing stress predictions for real-time adaptive VR interventions.

5. **Model Evaluation and Validation:** To evaluate the accuracy, robustness, and generalization ability of the proposed stress inference model using benchmark datasets (e.g., WESAD) and additional real-world physiological data.

3 METHODOLOGY

This component implements a temporally aware, multi-level stress inference engine to predict four stress levels (Relaxed, Mild, Moderate, High) through the analysis of HRV-derived features while it operates to eliminate the effects of normal human body hourly patterns. The methodology integrates Cosinor rhythmometry to estimate a user-specific 24-hour baseline (MESOR, amplitude, acrophase) [7] and it does circadian detrending using Cosinor residuals to isolate short-term stress deviations [3] and machine learning classifiers which use XGBoost as baseline and CNN-LSTM/LSTM sequence models and recent HRV stress detection work [5], [11] and deep learning evidence in affective computing [2] as training data. The researchers tested system performance through two methods which include cross-dataset validation and ablation tests because they found system performance to be unstable when used across different testing conditions [4]. Researchers used WESAD [1] as their main dataset for training and evaluation while they used extra occupational and longitudinal datasets to test time-based modeling and system reliability.

3.1 Research approach and justification

The research uses Design Science Research (DSR) methodology together with controlled experimental testing for its evaluation process.

The research utilizes Design Science Research (DSR) methodology together with controlled experimental testing. The research output produces an engineered artifact which uses Design Science Research (DSR) methodology because it developed a virtual reality system that employs machine learning and rhythmometry to create stress inference signals at different time intervals. The artifact solves two core problems which exist in stress detection systems. The first problem occurs when models achieve maximum performance on WESAD [1] benchmark testing yet their results do not extend to different datasets and sensors and user groups [4]. Deep learning stress inference methods are diverse and rapidly evolving. The authors of the IEEE survey of deep learning stress detection methods created architectural comparison standards which direct their evaluation work [2].

The experimental evaluation component measures whether the research tests circadian detrending together with personalized approaches results in better multi-level stress inference. The evidence demonstrates that HRV stress assessment accuracy improves when researchers eliminate circadian rhythm components from HRV [3]. Research demonstrates that multi-class HRV deep learning systems can effectively identify multiple HRV stress states which support multi-class stress modeling [5]. Multi-level classification problems which include demographic sensitivity and multilevel interpretability get strengthened through research about physiological multilevel classification systems. The reliability of wearable HRV devices throughout extended periods establishes the need for personalized solutions which enable baseline learning to occur in real world environments.

3.2 Technical Methods and algorithms

3.2.1 HRV feature extraction (time-domain and RR-series)

Input: This component receives the HRV as the input

For each window W containing N RR intervals:

- Mean RR

$$\overline{RR} = \frac{1}{N}\sum_{i=1}^{N} R\,R_i$$

- SDNN

$$SDNN = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(RR_i - \overline{RR}\right)^2}$$

- RMSSD

$$RMSSD = \sqrt{\frac{1}{N-1}\sum_{i=2}^{N}(RR_i - RR_{i-1})^2}$$

These HRV descriptors are used because multi-class HRV stress studies and deep learning pipelines rely on robust HRV features as stress-sensitive biomarkers [5], and multi-modal stress modeling explicitly integrates multiple HRV domains [9].

3.2.2. Frequency-domain HRV (LF/HF)

RR intervals are resampled to a uniform grid (interpolation method unspecified) and a PSD estimate is computed (e.g., Welch; PSD configuration unspecified). Frequency-domain features include: - LF power P_{LF} - HF power P_{HF} - LF/HF ratio:

$$\text{LF/HF} = \frac{P_{LF}}{P_{HF}}$$

Integration of time- and frequency-domain features is consistent with multi-modal deep stress detection practices that emphasize complementary feature domains [9].

3.2.3 Single-component Cosinor rhythmometry and parameter estimation

To model expected circadian HRV, a single-component Cosinor model is fitted per user to longitudinal HRV markers (primary: RMSSD; secondary: SDNN). The mathematical foundation follows Cosinor rhythmometry principles [7].

Cosinor model

$$Y(t) = M + A\cos\left(\frac{2\pi t}{\tau} + \phi\right) + \epsilon(t)$$

Where,

- M: MESOR
- A: amplitude
- ϕ: acrophase
- τ: period (fixed to 24 hours in the primary model)

Linearized form (regression estimation)
Let $\omega = \frac{2\pi}{\tau}$. The model is linearized:

$$Y(t) = M + \beta_1\cos(\omega t) + \beta_2\sin(\omega t) + \epsilon(t)$$

Parameter recovery:

$$A = \sqrt{\beta_1^2 + \beta_2^2}, \qquad \phi = \text{atan2}(-\beta_2, \beta_1)$$

3.2.4 Circadian detrending via Cosinor residuals

At inference time, the Cosinor baseline predicts $\hat{Y}(t)$. The residual is:

$$r(t) = Y_{\text{obs}}(t) - \hat{Y}(t)$$

Residuals are integrated into the feature vector as a "detrended channel," motivated by evidence that removing circadian effects from HRV can improve stress assessment consistency across times of day [3]. This directly targets the false-positive problem described in the literature (natural circadian HRV changes mistaken for stress) [3].

3.2.5 Personalization mechanisms

Personalization addresses inter-individual HRV variability and improves robustness, motivated by documented cross-dataset generalizability limitations [4] and evidence that wearable HRV is meaningful longitudinally [6].

User-specific Cosinor fitting. Each user has parameters (M_u, A_u, ϕ_u) learned from calibration/longitudinal data.

Normalization options (ablation-tested). - z-score normalization

$$z = \frac{x - \mu_u}{\sigma_u}$$

- Ratio normalization relative to predicted baseline

$$x_{\text{ratio}} = \frac{x}{\hat{Y}(t)}$$

- Residual-centric representation: use $r(t)$ plus compact HRV/time features (details unspecified and evaluated experimentally).

3.2.6 Machine Learning Architectures

The system uses this component to identify stress levels by analyzing bodily signals through its trained machine learning systems which function with HRV data extracted from Component A. The model aims to categorize human physiological states into four distinct stress levels which include Relaxed Mild Stress Moderate Stress and High Stress. The research shows that stress exists as a graded physiological response which requires multi-level classification instead of using binary classification [5] [10].

Two different model families are being tested to determine which one can accurately infer stress levels. The first model XGBoost functions as a gradient-boosted decision tree algorithm which achieves high accuracy when applied to structured physiological datasets. The XGBoost model establishes itself as the baseline because it shows strong performance while allowing researchers to understand how HRV features interact with stress levels.

3.2.7 Model Training Strategy

The supervised learning methods use labeled data from physiological stress datasets to train classification models. The training process uses weighted loss functions because physiological stress datasets show class imbalance with relaxed states occurring more frequently than stress

events. The model training process uses weighted categorical cross-entropy loss function to ensure proper representation of minority stress levels.

The training process uses data augmentation and transfer learning methods to enhance model generalization. HRV feature augmentation methods based on heart rate signal transformations are explored to simulate physiological variability and sensor noise conditions [8]. The strategy improves model robustness when exposed to actual sensing conditions found in real-world environments.

The WESAD dataset serves as the primary data source for initial model training because it functions as a standard benchmark dataset that researchers use to study wearable stress detection technology [1]. The researchers applied model fine-tuning with actual physiological data which they obtained from their wearable sensing system to solve the generalization problems which earlier research identified. The calibration step enables the model to learn individual physiological traits which assists in preventing performance decline across different datasets that previous HRV-based stress detection research has shown [4].

The second model family contains deep learning sequence models which include Long Short-Term Memory (LSTM) networks and hybrid CNN–LSTM models. LSTM networks function as effective tools for capturing temporal relationships between sequential HRV windows which enables researchers to use them for modeling time-varying physiological signals. The CNN–LSTM model uses convolutional layers to capture short-term time patterns which it passes to LSTM layers that perform extended sequence analysis. Previous research has demonstrated that CNN-based architectures can effectively detect stress patterns in HRV signals [11], while deep neural network approaches have shown promising results for multi-class stress classification [5]. The selection of these models is also consistent with current trends in affective computing and physiological stress detection research [2].

3.2.8. Model Optimization and Parameter Selection

The system uses cross-validation methods to optimize hyperparameters which guarantee machine learning and deep learning models will perform reliably. The model uses multiple training data subsets for evaluation through cross-validation which decreases the likelihood of overfitting. The system uses automated optimization techniques which include Bayesian optimization to conduct hyperparameter tuning by searching the parameter space to find the best model settings. The XGBoost model requires four main parameters which include tree depth learning rate number of estimators and regularization terms. The optimization process for deep learning models requires tuning multiple parameters which include the number of convolutional filters LSTM hidden units dropout rates and learning rates.

This systematic optimization process ensures that each model is evaluated under comparable conditions while maintaining reproducibility in the experimental design.

3.2.10 Real-Time Inference Considerations

This component needs to process physiological data with almost immediate results because it functions inside the closed-loop VR meditation system. The evaluation process uses trained model computational efficiency as a factor for assessment. The system architecture uses optimized inference formats to decrease computational requirements after the models undergo transformation from their trained state. The evaluation of the inference engine performance tests its capabilities through latency assessments and throughput measurements which verify its ability to work with the VR adaptation component. The system needs to maintain inference times between two specific points which depend on its system integration requirements to achieve real-time biofeedback functionality.

3.3 Tools, libraries, platforms

Category	Tool / Platform	Justification
Programming Environment	Python	Used as the core research environment due to its extensive ecosystem for machine learning, signal processing, and scientific computing.
Signal Processing & HRV Analysis	NeuroKit2	Provides reliable utilities for HRV computation and physiological signal processing, reducing implementation errors during HRV feature extraction.
Mathematical Rhythm Modeling	CosinorPy	Implements Cosinor rhythmometry for fitting periodic regression models to physiological signals and estimating MESOR, Amplitude, and Acrophase parameters.
Deep Learning Framework	PyTorch	Used to design and train deep learning architectures such as LSTM and CNN-LSTM networks with GPU acceleration support.
Machine Learning Model	XGBoost	Serves as the baseline classifier for structured HRV features, offering strong predictive performance and feature importance interpretability.
Machine Learning Utilities	Scikit-learn	Supports preprocessing, cross-validation, hyperparameter tuning, and evaluation metrics such as macro-F1 score and confusion matrices.
Deployment Environment	Docker	Provides containerized deployment to ensure reproducibility and consistency across development and testing environments.
Data Communication Protocol	MQTT / REST APIs	Enables reliable real-time communication between Component A and Component B for transmitting physiological data streams.

Time-Series Data Storage	InfluxDB	Optimized for storing time-stamped physiological data, enabling efficient querying and longitudinal data analysis.

Table 2: Tools and Platforms Used in the Component

3.4 Data Collection and Ethical considerations

3.4.1 Primary Datasets

The research will use multiple public physiological stress datasets to build and test their stress inference model. The WESAD dataset will serve as the primary benchmark dataset for model training and baseline evaluation due to its multimodal physiological recordings and widely accepted use in stress detection research [1].

The SWELL-KW dataset will enable cross-dataset validation to enhance model performance in various environmental conditions. The dataset includes office-based task stress responses which were collected under time pressure and interruption conditions, thus enabling the model to be tested under multiple stress scenarios [12].

The TILES-2018 dataset will support studies that investigate how physiological monitoring over time will enable personalized care. The dataset includes extended wearable sensor data which researchers collected from people in everyday situations, making it valuable for studying how different people show baseline physiological differences [13].

3.4.2. IOT Data Collection Plan

The HRV features which are derived from IOT device components will be used primarily for model validation and personalization experiments, allowing the system to be evaluated under realistic sensing conditions.

3.4.3 Ethical Considerations

All data collection procedures involving human participants will follow established ethical guidelines. Participants will give informed consent before participating in the study and researchers will explain the research purpose and the nature of the collected physiological data to them. All collected physiological data will be pseudonymized to protect participant privacy and researchers will keep identifying information in separate storage from physiological records. The dataset will be accessible only to research personnel who have been authorized to access it. Participants have the right to leave the study at any moment without facing any consequences. Every virtual reality experimental session will feature safety controls that enable participants to terminate the session when they experience discomfort. Researchers will obtain ethical approval from the institutional review board before they begin data collection efforts.

3.5 Validation Strategy and Performance Evaluation

3.5.1 Evaluation Protocol

The evaluation of the proposed stress inference model will use both within-dataset testing and cross-dataset testing methods. The evaluation within the dataset will employ k-fold cross-validation and leave-one-subject-out testing methods to determine the model's ability to generalize to new users. The cross-dataset evaluation will involve training the model with the WESAD dataset while testing it on other datasets which include SWELL-KW and the collected wearable pilot dataset. The evaluation method used in this study originates from earlier research which discovered that HRV-based stress detection models experience performance declines when used with different datasets and sensing configurations [4].

3.5.2 Performance Metrics

The proposed model will be evaluated using several classification performance metrics. The evaluation includes precision, recall and F1-score metrics which measure performance across all stress levels and macro-averaged F1-score which provides a complete performance assessment of all stress categories.

A confusion matrix will also be used to analyze the model’s ability to distinguish between different stress levels. The assessment will include receiver operating characteristic (ROC) curves and area under the curve (AUC) metrics which provide additional performance insights. The system requires real-time stress monitoring capabilities therefore its computational performance will be assessed through inference latency measurements and system throughput evaluation.

3.5.3 Ablation Studies

To better understand the contribution of each component of the proposed framework, several ablation experiments will be conducted. These experiments will evaluate the impact of circadian rhythm detrending, personalization mechanisms, and different feature sets on the overall model performance.Comparisons will also be performed between different model architectures, including XGBoost, LSTM, and CNN-LSTM networks, to identify the most effective approach for multi-level stress classification.

3.5.4 Success Criteria

The primary success criterion of this study is achieving a macro F1-score of at least 0.85 for the four-level stress classification task under controlled evaluation conditions. This target is informed by previous studies demonstrating strong performance of deep learning models for HRV-based stress detection [5]. Because cross-dataset evaluations lead to domain shift through differences in sensors and environments and participant populations, we will report performance variations across datasets instead of using a single threshold for analysis.

4 HIGH LEVEL SYSTEM ARCHITECTURE

4.1 Overview of proposed solution and its components

Input Layer

The system starts by gathering two primary forms of input data which include physiological signals and voice/audio recordings. Wearable sensors capture physiological data by measuring Photoplethysmography (PPG) and Electrodermal Activity (EDA) and skin temperature and motion through an accelerometer. The signals create the physiological basis which scientists require to conduct heart rate variability (HRV) research. The device microphone captures the voice data which scientists need for their research. The microphone input serves two purposes: it helps assess ambient noise and it enables voice-based stress analysis through user voice recording before and after the meditation session.

Physiological Intelligence Layer

The system processes physiological signals after data collection to obtain trustworthy and precise results. The component A system assesses the reliability of collected physiological signals by dividing the data into three categories which include reliable signals and recoverable signals and unreliable signals. Recoverable segments allow reconstruction through selective reconstruction methods while the system discards unreliable segments because they would lead to incorrect analysis results. Component B receives the validated signals which will undergo HRV extraction and feature computation processes. The HRV features get combined with time-aware circadian baseline modeling to enhance understanding of the user's physiological state which also improves stress level assessment compared to using raw HRV values.

The Session Validation Layer

Component D of the system performs two tasks. The first task requires the system to check ambient noise levels to find out whether the user can meditate in their current environment. The system conducts voice sampling at two points during the meditation session to collect audio data which it evaluates through stress assessment methods that use vocal analysis. The system generates voice-based stress assessments which the system compares to HRV-based stress estimates that the physiological analysis module produces. Anomaly detection mechanisms use detection of atypical patterns to mark sessions as either ineffective or irregular which helps improve the accuracy of system outputs

The Adaptation Layer

System Component C functions as the engine which enables users to personalize their experience of the system while its adaptation system delivers tailored content to users. The system uses stress level data which it obtains from monitoring physiological signals together with confidence scores from the signals and results from cross-modal validation and data from prior sessions to create a dynamic VR meditation environment. The personalization module determines how to adapt different environmental parameters such as scene type, lighting conditions, background soundscape, and weather effects. The system uses these changes to create a meditation experience which matches the user's current mental and physical condition to achieve better results in stress relief.

Output Layer

The system's last layer presents personalized meditation content to users while delivering their meditation performance feedback. The virtual reality environment adapts throughout the session based on personalization module adaptive decisions which help users achieve better relaxation and stress relief results. The system produces a session-level report at session termination which contains user stress change data and assesses trustworthiness based on signal quality and cross-modal validation results. The output displays user stress patterns which help users monitor their progress during meditation practice.

4.2 System diagram of components

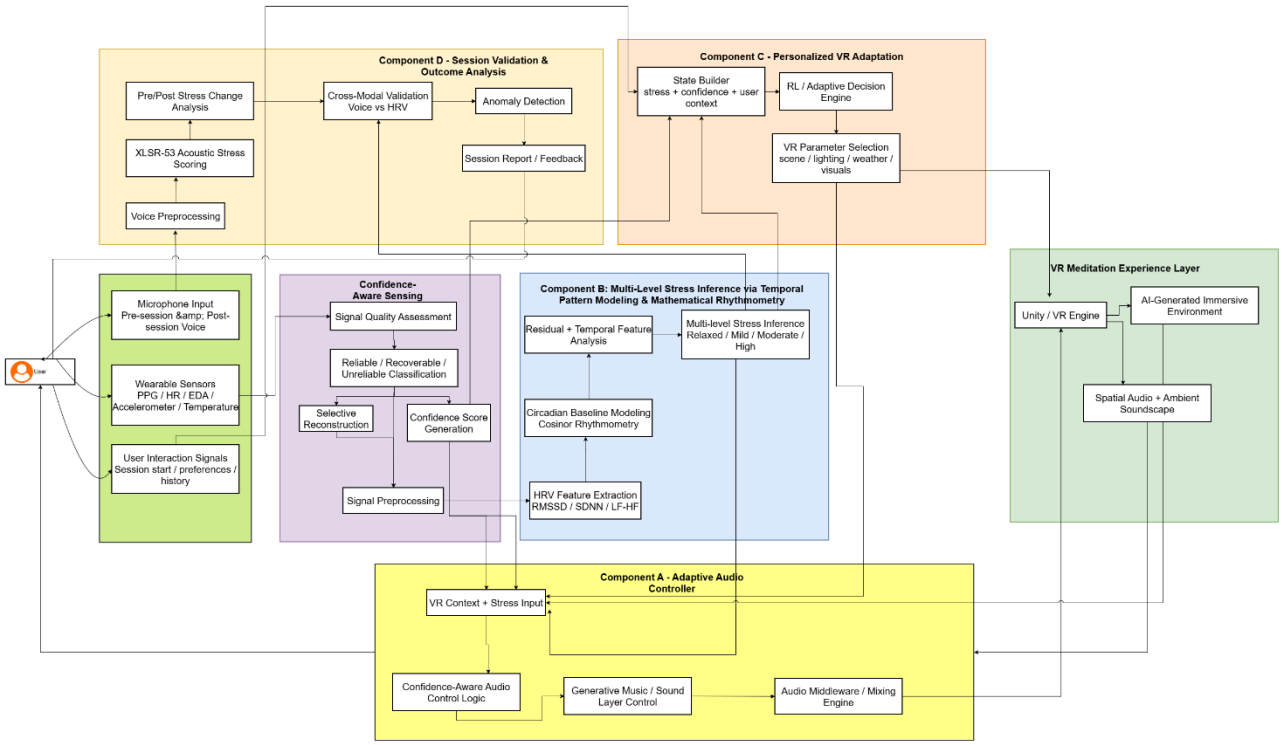


Figure 1: System diagram of components

4.3 Temporally aware multi-level HRV-based stress inference model and Its Sub-Modules

The component works as the intelligence layer of the proposed system to convert heart rate data into psychological stress predictions. Component A handles the acquisition and preprocessing of physiological signals while Temporally aware multi-level HRV-based stress inference model handles feature extraction and circadian rhythm analysis and stress detection.

This component contains three primary functional submodules which make up its internal structure. The HRV Feature Derivation Layer serves as the first submodule of the system. The module receives a cleaned RR interval stream which Component A generated. The RR intervals are used to calculate standard Heart Rate Variability (HRV) metrics which include RMSSD and SDNN and the LF/HF ratio. The features track how the autonomic nervous system functions and they show how the body responds to stress. The system processes HRV features through time window organization which combines them with contextual data that includes time-of-day information.

The Circadian Rhythm Modeling Layer functions as the main innovative element of the system. The module uses Cosinor rhythmometry to determine the normal physiological HRV baseline

which should occur during a complete 24-hour circadian rhythm cycle. The Cosinor model produces three key parameters: MESOR, which represents the rhythm-adjusted mean; Amplitude, which describes the magnitude of circadian variation; and Acrophase, which indicates the time of peak physiological activity. The model uses these parameters to forecast the HRV value which will occur at specific times throughout the day. The system calculates the difference between actual HRV features and predicted circadian baseline values to create a residual signal which indicates psychological stress levels that exceed normal biological variations.

The third sub-module is the Machine Learning Stress Inference Engine. This module uses the residual features together with the original HRV feature vector to forecast the user's stress level. The researchers investigate multiple machine learning models which include XGBoost Long Short-Term Memory LSTM networks and hybrid CNN–LSTM architectures. The models use both current HRV patterns and past HRV patterns to determine the physiological state which has four categories that include relaxed, mild stress, moderate stress and high stress.

4.4 Component Interactions

The system has three components which include Component B as its central intelligence layer which connects with the upstream sensing module and downstream adaptive response module. The system uses Component A to convert physiological data into stress predictions that the system can utilize.

Component A functions as the starting point of the interaction because it handles both physiological signal acquisition and signal-level preprocessing functions. The device uses Component A to gather raw ECG or PPG signals from either a wearable device or an IoT device, before it proceeds to eliminate all noise and motion artifacts, and finally it detects RR intervals. Component B receives the cleaned RR interval stream from Component A. Component A currently functions to obtain accurate RR interval data which it will distribute for advanced data analysis; it currently lacks the capacity to predict stress or model physiological functions.

Component B uses the received RR intervals to calculate HRV features which include RMSSD, SDNN, and LF/HF ratio; these features are combined with information about the current time of day. The system uses circadian rhythm modeling methods to process these features, and Cosinor analysis is used to estimate the expected physiological baseline for the user. The machine learning inference engine receives both the residual signals and HRV feature vectors, which it uses to determine the current stress level in four categories: relaxed, mild stress, moderate stress, and high stress.

The output of Component B is sent to Component C, which handles adaptive intervention. Component C uses the predicted stress level to change the VR meditation environment in real time. When the system detects a high stress state, Component C will display a peaceful visual scene and

provide slower breathing guidance and softer audio feedback. Component B delivers the decision signal which enables VR adaptation to occur in real time.

Component B can interact with a subsystem that stores historical data and user profiles, which contains actual physiological data and circadian model parameters used for creating personalized experiences and updating baseline information. The inference engine improves its comprehension of user physiological behavior through these ongoing interactions. Component B establishes a connection between raw physiological data and intelligent stress assessment which leads to adaptive system response, thus serving as an essential link between data collection and user-oriented intervention.

4.5 Data Flow Explanation

The data flow diagram of the proposed component is as follows.

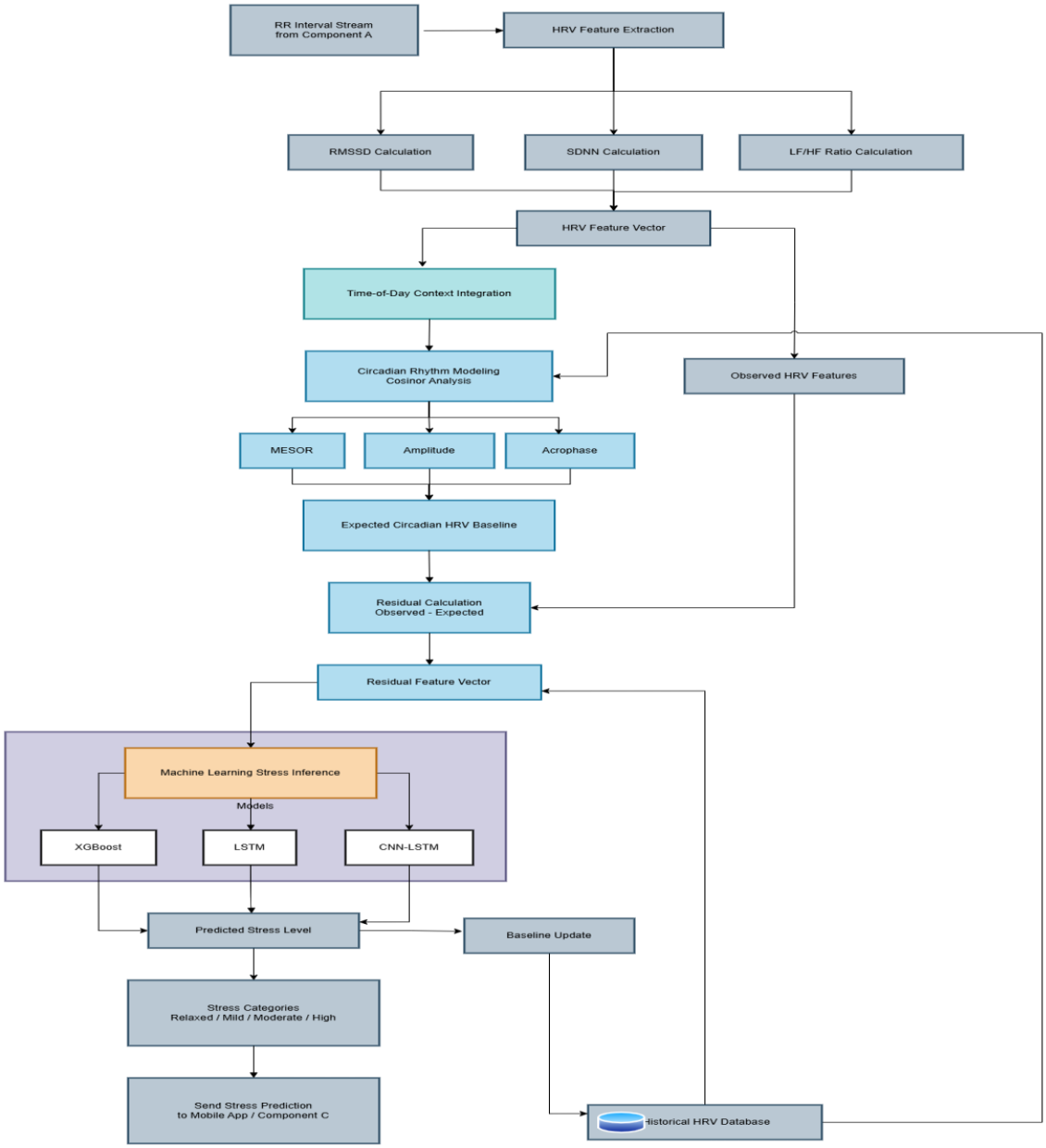


Figure 2: Data flow diagram

The system starts its data processing by using Component A to collect biological data through its signal acquisition system. The wearable sensors gather unprocessed cardiac data which undergoes cleaning procedures to obtain RR intervals. The cleaned RR interval stream goes to Temporally aware multi-level HRV-based stress inference model after it has been processed.

Temporally aware multi-level HRV-based stress inference model uses RR interval data to calculate HRV features which it combines with time-of-day information. The circadian modeling module then estimates the expected physiological baseline using Cosinor rhythmometry. The system calculates residual signals which show abnormal circadian patterns by comparing actual HRV data with predicted HRV data from circadian rhythms. T

The machine learning stress inference engine receives two types of input which include residual features and original HRV features. The inference engine processes these features and produces a probabilistic prediction of the user's stress state across four stress levels. The system transmits the predicted stress level to its higher-level components which include the VR adaptation module that modifies the user environment based on the stress level.

4.4 Scalability and Security considerations

The proposed architectural design enables the simultaneous processing of scaling physiological data while maintaining secure protection of confidential user details. The system needs to process increasing amounts of time-series data because wearable devices produce continuous streams of physiological signals. The architecture solves this problem through its modular design which uses individual components to handle distinct tasks in the processing sequence. The system uses a modular structure which enables each component to expand its capacity without affecting other parts of the system. This component includes a stress inference engine which functions as an independent processing service that enables the system to handle multiple users through automatic resource distribution.

The system achieves scalability through its implementation of time-series databases which store physiological data and optimize performance for both high-frequency sensor data and smooth retrieval of extended health records. The system uses these databases to store past physiological data which helps in creating personalized experiences and establishing normal circadian patterns without disrupting ongoing stress inference operations. The system uses lightweight communication protocols such as MQTT and REST-based APIs to transmit physiological data streams between components while maintaining reliable performance across distributed devices.

Security and privacy require special attention during the process of handling physiological information and health-related data. The system therefore implements multiple protection measures which safeguard user data. All physiological data transmissions between components should occur through secure communication channels, such as encrypted network protocols, to prevent unauthorized interception. The system requires that all collected physiological data must

be protected through encryption methods and access control systems which permit only designated system elements and research personnel to access the data.

The system will use data pseudonymization to protect user identity because it stores personal identifiers in separate locations from physiological data. This approach reduces the risk of linking sensitive physiological data directly to individual users. The system needs access control policies which will limit system entry to authorized users and their approved applications.

The proposed architecture enables scalable deployment through its combination of modular system design and efficient data management methods and strong privacy protection systems which safeguard the security and confidentiality of physiological data.

5 USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed AI-based stress inference component involves several stakeholders who interact with or benefit from the system.

5.1.1 Primary Stakeholders

1. End Users (Meditation App Users):

These are individuals who use the VR meditation system to monitor and manage their stress levels. They benefit from personalized stress detection and adaptive meditation guidance.

2. Mental Health Practitioners:

Psychologists, therapists, and wellness coaches who may use the system to monitor physiological indicators of stress and recommend appropriate interventions.

3. Researchers and Data Scientists:

Researchers working in affective computing, bioinformatics, and digital health who analyze the physiological data and improve the stress inference models.

5.1.2 Secondary Stakeholders

1. System Developers / Engineers:

Software developers responsible for implementing the HRV processing pipeline, machine learning models, and cloud infrastructure.

2. Healthcare Institutions and Wellness Platforms:

Organizations that may deploy the system for employee wellness programs or clinical support tools.

3. Device Manufacturers:

Companies who produce wearable sensors (ECG/PPG devices) that provide physiological data to the system.

5.2 Functional Requirements

This component needs to execute specific functions according to its functional requirements.

1. The system shall receive processed physiological signals (HRV features derived from ECG or PPG data) from the data acquisition component.
2. The system shall perform HRV feature extraction including metrics such as RMSSD, SDNN, and LF/HF ratio.
3. The system shall implement circadian rhythm modeling using Cosinor or similar mathematical rhythmometry techniques.
4. The system will create an ongoing physiological baseline which changes according to the biological rhythms of each user.
5. The system will calculate HRV residual values by measuring actual HRV against established circadian base values.
6. The system will feed HRV data together with time-based residual data to a machine learning classification system.
7. The system will identify 4 different stress levels which include Relaxed, Mild Stress Moderate Stress, High Stress.
8. The system will send the detected stress level to the VR meditation system which will use it to modify the meditation process.
9. The system will maintain an ongoing process of personalized baseline updates whenever new physiological measurements are obtained.
10. The system will keep all processed physiological data and predictive results in secure storage within the cloud database so they can be analyzed later.

5.3 Non- functional Requirements

The system's non-functional requirements describe its essential system quality attributes.

1. Security:
   The system will protect all physiological information by encrypting it during both data transfer and data storage processes. The system will protect user privacy by converting all personal identification information into non-identifiable formats. The system will use authentication methods together with access control systems to protect its resources.
2. Performance:

The system will conduct stress inference processes throughout the day while maintaining low system impact during its daily operations. The system will process physiological data streams with minimal latency. The system will provide continuous monitoring capability without creating major system performance demands.

3. Scalability:
   The system will handle multiple users at the same time because it uses cloud infrastructure. The system architecture allows for simple attachment of new sensors plus additional data sources.
4. Reliability:
   The system will function continuously with stable performance while it monitors physiological changes. The stress inference model will produce identical results for all users throughout various time intervals.
5. Usability:
   The system will produce stress level outputs which users can easily understand. The system will function properly because it requires no technical knowledge from its users.
6. Maintainability:
   The system architecture will permit fast implementation of updates for all machine learning models plus their corresponding algorithms. The system will enable modular system upgrades which will help organizations improve their future stress detection capabilities.

6 COMMERCIALIZATION PLAN

6.1 Target market and customer persona

The AI-Based Adaptive VR Meditation System has been designed to provide meditation solutions suitable for all users who exist in both consumer and professional environments. The primary target market includes individual users seeking effective stress relief and guided meditation, corporate wellness programs aiming to improve employee well-being, and healthcare providers such as mental health practitioners, therapy centers, and hospitals that require advanced tools for monitoring and managing stress. In addition, universities and research institutions interested in affective computing and biofeedback systems represent an important segment.

The secondary market includes VR wellness studios and meditation coaches and health-tech startups and wearable device manufacturers who can use the system to enhance their current platforms. The typical customer persona includes working professionals who experience daily stress and need customized relaxation solutions or healthcare providers who want dependable tools to track patient stress levels.

6.2 Value Proposition

The meditation system provides its unique value because it delivers complete meditation solutions through its system which integrates physiological monitoring together with immediate stress detection and virtual reality systems and custom therapeutic audio and session assessment. The system creates personalized stress relief results through its continuous adjustment to user physiological changes which distinguishes it from conventional meditation applications that deliver fixed content. The system establishes its clinical value through its ability to deliver measurable results which build trust among users.

6.3 Revenue Model

The system uses a hybrid revenue model that integrates Business-to-Consumer (B2C) and Business-to-Business (B2B) revenue streams. Users can access the platform through subscription-based pricing while organizations such as clinics and corporate wellness providers and research institutions can purchase enterprise licenses. The system offers commercial potential through its Software Development Kits (SDKs) and APIs which enable third-party developers to incorporate stress inference and adaptive VR elements into their applications. Additional revenue streams may include hardware bundle partnerships with wearable device manufacturers and research deployment packages.

6.4 Cost Estimation

The main expenses of the system consists of three components which are software development costs and cloud infrastructure expenses and system integration fees. Development costs involve building machine learning models, VR environments, and backend systems. The expenses related to cloud services include three types of costs which are data storage fees and real-time processing charges and secure communication expenses. The expenses for hardware are low because the system only needs wearable sensors and VR devices which can be obtained through partnerships. The ongoing expenses for operations include maintenance costs and update expenses and technical support fees.

6.5 Pricing Strategy

The pricing strategy is designed to accommodate different user segments. The individual user segment requires a low-cost monthly or annual subscription model which will drive user adoption. Enterprise and clinical users can choose between two pricing options which include user volume-based pricing and annual licensing agreements. Developers and partners can access API or SDK services through two different pricing models which include usage-based and subscription-based options. The tiered pricing system enables businesses to operate at various market levels while maintaining both flexible and scalable operations.

6.6 Competitive Advantage

The system's primary competitive strength emerges from its complete integrated system which can adapt to different requirements. The system merges multiple intelligent systems which include physiological monitoring and multi-tiered stress assessment and interactive virtual reality customization and voice-powered session authentication and dynamic sound management into a unified platform which differs from traditional meditation applications. The system provides real-time customization which enhances treatment precision and produces detectable healing results. The system solves major problems between existing solutions which include their inability to create personalized experiences and understand user behavior and they lack proper authentication methods.

6.7 Intellectual Property Considerations

The system includes several elements that can be protected as intellectual property. The system presents a new patentable invention that combines circadian rhythm modeling with machine learning to create a method for detecting stress. The developed algorithms and system architecture together with their software components receive protection through licensing agreements and copyright law. The organization must establish effective data protection methods to meet privacy regulations when it manages sensitive physiological information. By protecting both the technical methods and the implementation, the long-term commercial value of Component B can be maintained and strengthened.

7 BUDGET AND JUSTIFICATION

7.1 Budget Table

Category	Item Description	Estimated Cost (LKR)
Hardware Costs	PPG wearable device / sensor (basic)	15,000
	Existing laptop/workstation	0
Software Licenses	Open-source tools (Python, TensorFlow, etc.)	0
	Optional tools (MATLAB – academic license)	0
Cloud Services	AWS Free Tier (development phase)	0
	Additional cloud usage (testing & deployment)	5,000
Human Effort Cost	Development effort (estimated academic contribution)	50,000
Miscellaneous	Internet, utilities, minor expenses	3,000
Total Estimated Cost		**73,000 LKR**

Table 3: Budget Table

7.2 Justification

Component B has a budget that remains low because its primary components function as software applications. The project incurs one hardware expense which consists of a basic wearable device that functions to gather physiological measurements. The existing personal laptop allows all development work to proceed without needing extra computing hardware.

The project incurs low software expenses because it depends on open-source software which includes Python and machine learning libraries. The university licenses provide access to optional tools like MATLAB which users can access without incurring additional expenses.

The project uses cloud services for its main purposes of testing and deployment. The development process needs free-tier services as the primary resource but the project includes a small budget to support extended usage and performance testing. The system operates in real-time processing mode whenever necessary.

Human effort cost constitutes the largest budget component because it requires extensive time and specialized knowledge to create and test the stress inference model. The academic project requires no actual financial costs, yet it demonstrates the true academic worth of all work performed.

The budget shows that Component B functions as a cost-efficient solution which maintains its ability to grow and operate without requiring major financial expenses for development and implementation.

8 WORK BREAKDOWN STRUCTURE (WBS)

8.1 Work Breakdown Structure (WBS)

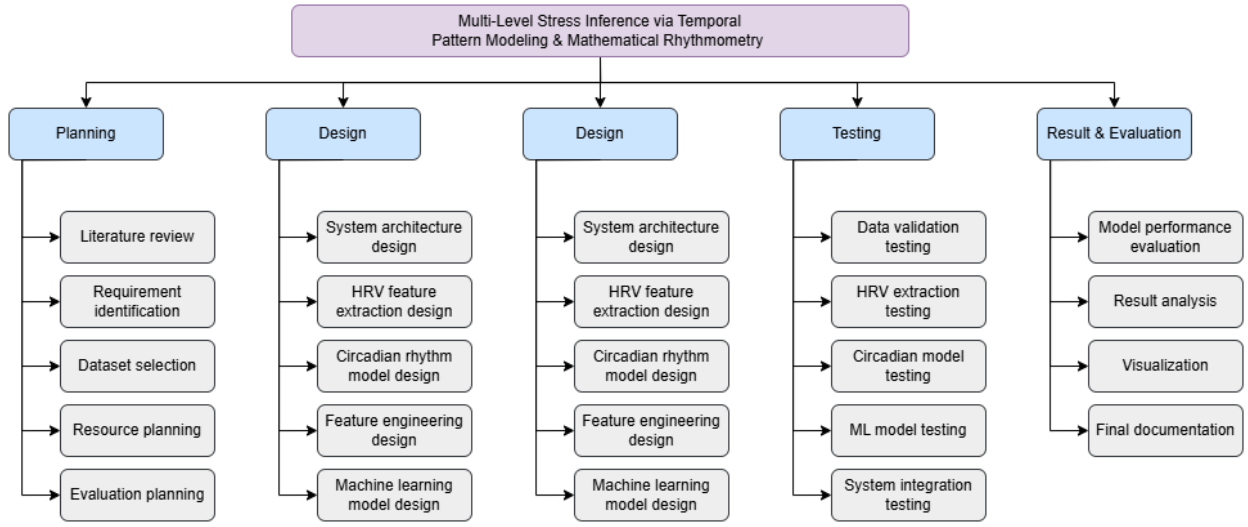


Figure 3: Work Breakdown Structure (WBS)

8.2 Timeline

Task	Start Date	End Date	Duration
Feasibility Study			
Initial Research Idea Exploration	Nov 2025	Nov 2025	2 weeks
Preliminary Literature Review	Nov 2025	Nov 2025	2 weeks
Identify Research Gap	Nov 2025	Nov 2025	1 week
Requirement Gathering & Background Study			
Detailed Literature Review	Dec 2025	Jan 2026	4 weeks
Study of HRV Stress Detection Methods	Dec 2025	Jan 2026	3 weeks
Study of Circadian Rhythm Modeling	Dec 2025	Jan 2026	3 weeks
Topic Approval Phase			
Topic Assessment Form (TAF) Submission	Jan 2026	Feb 2026	2 weeks
Proposal Preparation			
Proposal Writing	Feb 2026	Mar 2026	4 weeks
Proposal Submission	Mar 2026	Mar 2026	1 day
Proposal Presentation Preparation	Mar 2026	Mar 2026	1 week
Proposal Presentation	Mar 2026	Mar 2026	3 days
System Design			

High-Level System Architecture Design	Mar 2026	Apr 2026	3 weeks
Component Design (IoT, AI, VR Modules)	Mar 2026	Apr 2026	3 weeks
Dataset Preparation and Experiment Planning	Apr 2026	Apr 2026	2 weeks
Development Phase			
HRV Feature Extraction Development	Apr 2026	May 2026	4 weeks
Circadian Rhythm Modeling (Cosinor)	Apr 2026	May 2026	4 weeks
Stress Classification Model Development	May 2026	Jun 2026	4 weeks
Progress Evaluation Phase			
Checklist Submission Preparation	Apr 2026	May 2026	1 week
Progress Presentation 1	May 2026	May 2026	1 day
Model Training & Experimentation			
Model Training & Hyperparameter Tuning	Jun 2026	Jul 2026	5 weeks
Cross-Dataset Evaluation	Jul 2026	Jul 2026	3 weeks
Real-Time System Integration	Jul 2026	Aug 2026	4 weeks
Progress Evaluation Phase 2			
Progress Presentation 2 Preparation	Aug 2026	Aug 2026	1 week
Progress Presentation 2	Aug 2026	Aug 2026	3 days
Final System Development			
System Optimization and Testing	Aug 2026	Sep 2026	4 weeks
Result Analysis and Documentation	Sep 2026	Sep 2026	3 weeks
Final Submission Phase			
Final Report Writing	Sep 2026	Oct 2026	4 weeks
Website Development	Sep 2026	Oct 2026	3 weeks
Final Report Submission	Oct 2026	Oct 2026	1 day
Website Submission	Oct 2026	Oct 2026	1 day
Final Presentation and Viva	Oct 2026	Oct 2026	3 days
Website Evaluation & Logbook Submission	Oct 2026	Oct 2026	3 days
Research Paper Submission	Oct 2026	Oct 2026	1 day
Publication Phase			
Research Paper Publication Evidence Submission	Oct 2026	Dec 2026	Ongoing

Table 4: Project Timeline

9 GANTT CHART

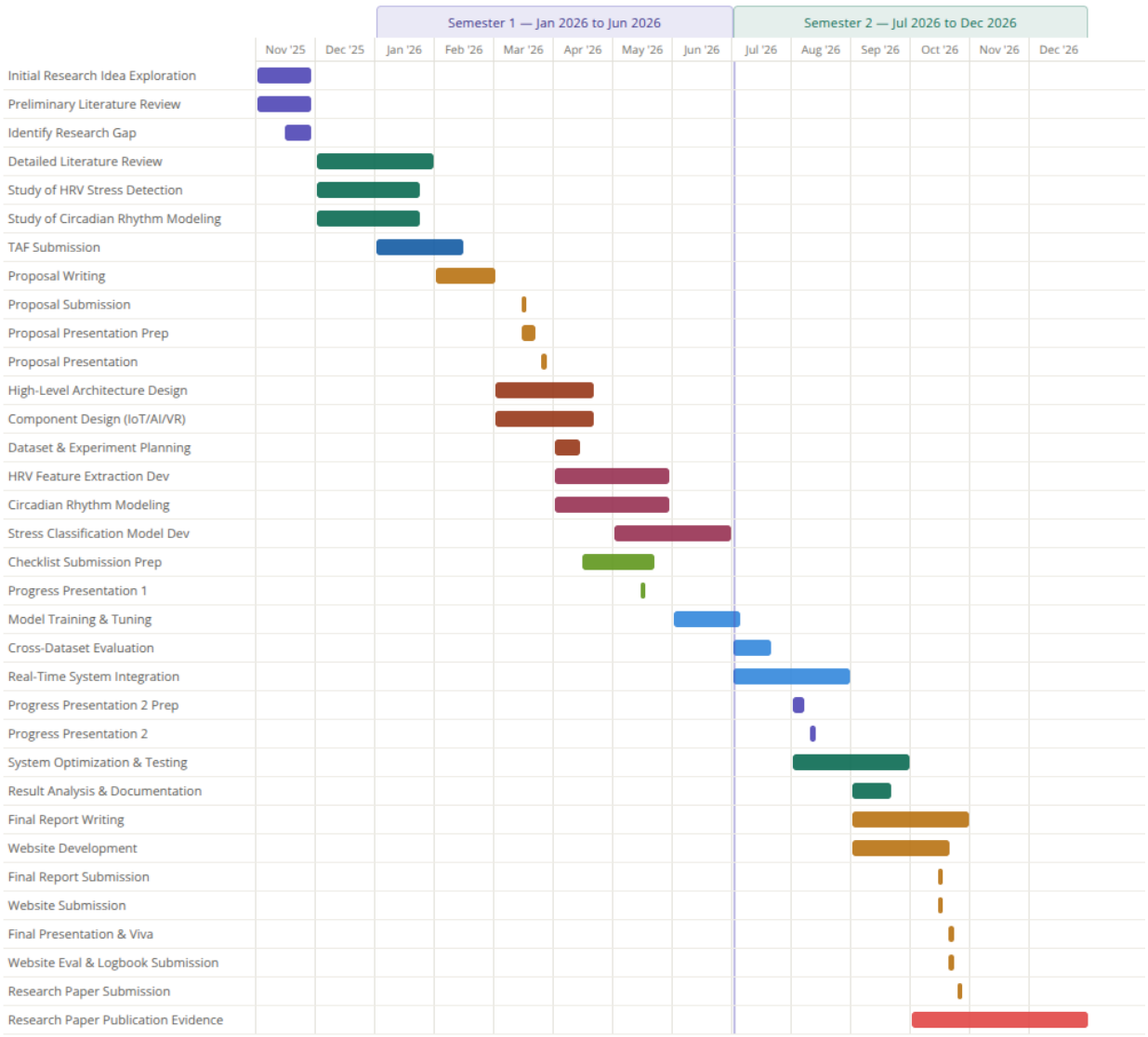


Figure 4: Gantt chart

10 REFERENCES LIST

[1] P. Schmidt, A. Reiss, R. Duerichen, C. Marberger, and K. Van Laerhoven, "Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection," in *Proceedings of the 20th ACM International Conference on Multimodal Interaction*, Boulder CO USA: ACM, Oct. 2018, pp. 400–408. doi: 10.1145/3242969.3242985.

[2] M. Kyrou, I. Kompatsiaris, and P. C. Petrantonakis, "Deep Learning Approaches for Stress Detection: A Survey," *IEEE Trans. Affect. Comput.*, vol. 16, no. 2, pp. 499–517, Apr. 2025, doi: 10.1109/TAFFC.2024.3455371.

[3] Y. Shen, Z. Fang, T. Zhang, F. Yu, Y. Xu, and L. Yang, "Heart rate variability with circadian rhythm removed achieved high accuracy for stress assessment across all times throughout the day," *Front. Physiol.*, vol. 16, p. 1535331, Apr. 2025, doi: 10.3389/fphys.2025.1535331.

[4] M. Benchekroun, P. E. Velmovitsky, D. Istrate, V. Zalc, P. P. Morita, and D. Lenne, "Cross Dataset Analysis for Generalizability of HRV-Based Stress Detection Models," *Sensors*, vol. 23, no. 4, p. 1807, Feb. 2023, doi: 10.3390/s23041807.

[5] J. A. Mortensen, M. E. Mollov, A. Chatterjee, D. Ghose, and F. Y. Li, "Multi-Class Stress Detection Through Heart Rate Variability: A Deep Neural Network Based Study," *IEEE Access*, vol. 11, pp. 57470–57480, 2023, doi: 10.1109/ACCESS.2023.3274478.

[6] R. Hernandez *et al.*, "Resting Heart Rate Variability Measured by Consumer Wearables and Its Associations with Diverse Health Domains in Five Longitudinal Studies," *Sensors*, vol. 25, no. 23, p. 7147, Nov. 2025, doi: 10.3390/s25237147.

[7] G. Cornelissen, "Cosinor-based rhythmometry," *Theor. Biol. Med. Model.*, vol. 11, no. 1, p. 16, Dec. 2014, doi: 10.1186/1742-4682-11-16.

[8] P.-J. Chen, W.-S. Chien, H.-Y. Chen, and C.-C. Lee, "Stress Detection Using HRV Features Augmentation Based on Heart Rate Signal Transformation," in *2024 IEEE 20th International Conference on Body Sensor Networks (BSN)*, Chicago, IL, USA: IEEE, Oct. 2024, pp. 1–4. doi: 10.1109/BSN63547.2024.10780687.

[9] J.-Z. Xiang, Q.-Y. Wang, Z.-B. Fang, J. A. Esquivel, and Z.-X. Su, "A multi-modal deep learning approach for stress detection using physiological signals: integrating time and frequency domain features," *Front. Physiol.*, vol. 16, p. 1584299, Apr. 2025, doi: 10.3389/fphys.2025.1584299.

[10] A. Hemakom, D. Atiwiwat, and P. Israsena, "ECG and EEG based detection and multilevel classification of stress using machine learning for specified genders: A preliminary study," *PLOS ONE*, vol. 18, no. 9, p. e0291070, Sep. 2023, doi: 10.1371/journal.pone.0291070.

[11] S. Quoy, D. Istrate, M. Benchekroun, and V. Zalc, "HRV Based Stress Detection Using Convolutional Neural Networks (CNNs)," in *2024 International Conference on Content-Based Multimedia Indexing (CBMI)*, Reykjavik, Iceland: IEEE, Sep. 2024, pp. 1–5. doi: 10.1109/CBMI62980.2024.10859222.

[12] S. Koldijk, M. Sappelli, S. Verberne, M. A. Neerincx, and W. Kraaij, "The SWELL Knowledge Work Dataset for Stress and User Modeling Research," in *Proceedings of the 16th*

International Conference on Multimodal Interaction, Istanbul Turkey: ACM, Nov. 2014, pp. 291–298. doi: 10.1145/2663204.2663257.

[13] K. Mundnich *et al.*, "TILES-2018, a longitudinal physiologic and behavioral data set of hospital workers," *Sci. Data*, vol. 7, no. 1, p. 354, Oct. 2020, doi: 10.1038/s41597-020-00655-3.

11 APPENDICES

11.1 Appendix A: Dataset Description and Input Structure

The following data is used for stress inference in Component B:

Dataset Details

- Dataset source: *(e.g., WESAD / collected dataset)*
- Signal type:
    - Heart Rate Variability (HRV)
    - RR interval sequences

Extracted Features

- RMSSD
- SDNN
- LF/HF ratio

Input Structure

- Time-series data using sliding windows
- Input format:
    - RR interval sequences or HRV feature sequences
- Example:
    - (time_steps × features)

Evaluation Inputs

- Stress level rating before session (1–5 scale)
- Stress level rating after session (1–5 scale)
- User perception of system accuracy

11.2 Appendix B: Detailed Diagrams

Some important diagrams are listed as follows.

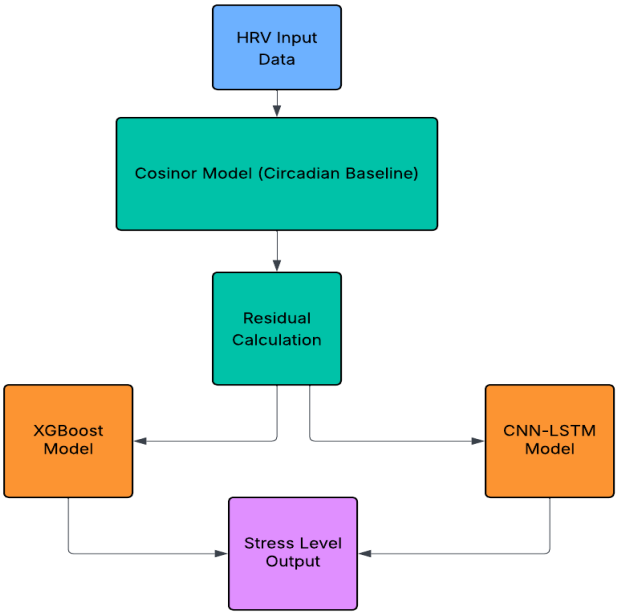


Figure 5: Architecture diagram of stress inference layer

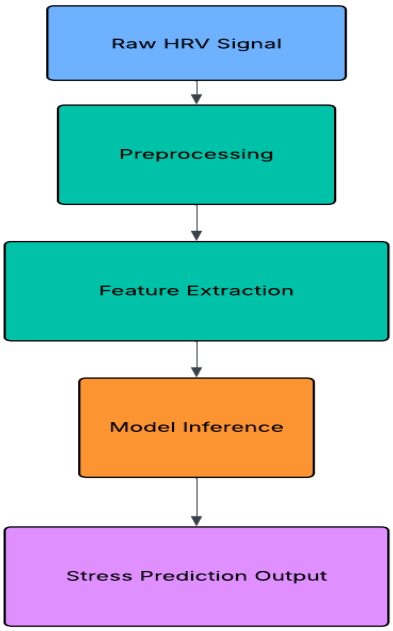


Figure 6: Data flow diagram of stress inference layer

11.3 Appendix C: Model Architecture Details

XGBoost Model

- Input features:
    - RMSSD
    - SDNN
    - LF/HF ratio
    - Cosinor residual
- Type: Multi-class classification
- Output:
    - Relaxed
    - Mild Stress
    - Moderate Stress
    - High Stress

CNN-LSTM Model

- Input:
    - Time-series HRV sequences
- Input shape:
    - (time_steps × features)

Architecture

- 1D CNN layer → Extract local patterns
- LSTM layer → Capture temporal dependencies
- Dense layer → Final classification

Purpose

- Learn temporal patterns in HRV signals
- Improve stress prediction accuracy

11.4 Appendix D: Risk Analysis

Risk	Impact	Mitigation
Noisy HRV data	Incorrect stress predictions	Apply preprocessing and filtering
Limited dataset	Poor model performance	Use public datasets / augmentation
Overfitting	Low generalization	Apply cross-validation
Incorrect circadian modeling	Misclassification	Validate Cosinor model
Real-time delay	Performance issues	Optimize model complexity

Table 5: Risk analysis

11.5 Appendix E: Ethical Considerations

Data Privacy

- HRV data is treated as sensitive.
- No personally identifiable information is stored
- Data anonymization is applied.

Consent (if data collection is performed)

- Participants are informed prior to data collection
- Participation is voluntary.

Safety Considerations

- System is intended for wellness support only
- No clinical diagnosis or medical decisions are provided

This component does not provide clinical diagnosis and is intended for wellness support only.